

1. Current Division Generalisation

Consider the given circuit with const. current source and ideal Resistors

All resistors are connected across same terminal A & B.

→ All branches have same voltages across them.

In i^{th} Branch, Potential drop is V_i

$$V_i = V_{i+1} \quad \forall i = 1 \text{ to } n-1$$

Applying ohm's law on i^{th} branch.

$$V_i = i_i R_i$$

$$\Rightarrow i_1 R_1 = i_2 R_2 = i_3 R_3 = \dots = i_n R_n = V$$

now looking at node A. By kirchoff current law

$$\sum \text{current at node} = 0$$

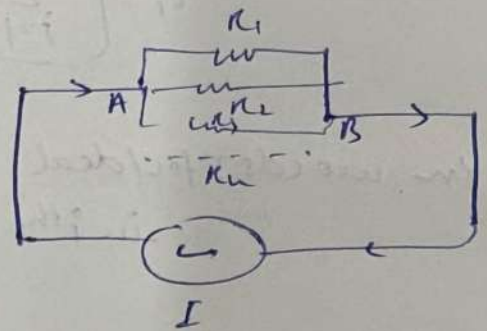
$$\text{so } I - \sum i_i = 0 \Rightarrow \sum i_i = I \quad \text{--- (1)}$$

now replace all these resistor with one eq. resistor having an equivalent resistance, across A and B same as these n resistor. So our circuit can be drawn as

$$V = R_{eq} I \quad \text{--- (2) Again by Ohm's law}$$

using (1) (2) (3)

$$R_{eq} = \left(\sum_{i=1}^n \frac{1}{R_i} \right)^{-1}$$



Current in i^{th} Branch is given by i , where

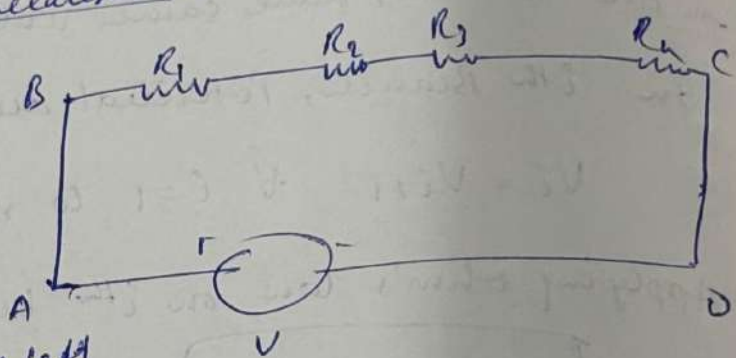
$$i_i R_i = V = R_{eq} I$$

$$i_i = \left[\frac{\sum_{j=1}^n \left(\frac{1}{R_j} \right)}{R_i} \right]^{-1} I$$

(in words) for ideal resistor in parallel circuit in i^{th} branch (equivalent Resistance) R_i

2. Voltage Division Generalisation

Given: A circuit with constant voltage source. Sourcing V volt in a circuit with n ideal resistor in series (i^{th} resistor from left has Resistance R_i)



Applying KVL and Ohms law on a closed loop ABCDA

$$\sum \text{voltage} = 0$$

$$V - iR_1 - iR_2 - \dots - iR_n = 0$$

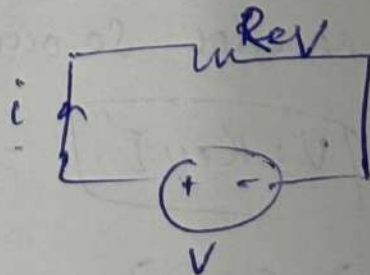
$$V = i \sum_{i=1}^n R_i$$

Hence as we consider equivalent of these n resistor to be R_{eq}

$$V = i R_{eq}$$

$$i R_{eq} = i \sum_{i=1}^n R_i$$

$$R_{eq} = \sum_{i=1}^n R_i$$



now $V = i \sum_{i=1}^n R_i \Rightarrow i = \frac{V}{\sum_{i=1}^n R_i}$

In general after applying Ohm's law

$$V_i = \frac{V R_i}{R_{eq}}$$

for n resistor in parallel

$$i_i = \frac{\left(\sum_{i=1}^n \frac{1}{R_i} \right)^{-1} V}{R_i}$$

$$V_i = V_{i+1} = V \quad \forall i = 1 \text{ to } n-1$$

for n resistor in series

$$i_i = i = \frac{V}{R_{eq}} \quad \forall i = 1 \text{ to } n$$

$$V_i = \frac{R_i V}{\sum_{j=1}^n R_j}$$